KisanVriddhi Platform Specification

End-to-End AI-Powered Agricultural Decision Support Intelligence

Author: Antigravity Coding Assistant	**Target:** Google AI Hackathon
Tech Stack: FastAPI, React, Gemini AI, gTTS	**Status:** Production Ready
Date: July 2026	**Vulnerability Code:** Safe / Non-Traversable

1. Executive Summary

KisanVriddhi is a comprehensive digital agritech ecosystem designed to help Indian farmers navigate changing weather risks and maximize crop outcomes through cutting-edge AI. Built to directly align with the core evaluation metrics of the **Google AI Hackathon**, the platform avoids displaying static spreadsheets, transforming real-world sensor streams and API data into fully personalized, multilingual action checklists. Through direct officer integration, the platform also scales state-wide interventions, allowing government agricultural extension agronomists to manage crop alerts, track outbreaks, and deploy target field visits seamlessly.

2. Core Platform Features

• Digital Farm Profile

The foundation of KisanVriddhi personalization. Captures soil type (Loamy, Clay, Alluvial, Black, Sandy), soil pH (0-14, supporting fine matching), land size (in acres), groundwater table depth, preferred local language (English, Hindi, Malayalam, Telugu), and crop history. This profile guarantees all advice is tailored to the exact physical attributes of the farmer's land.

• Farmer Intelligence Dashboard

A real-time overview hub showing active smart weather alerts, risk gauges (Weather, Water, Disease, and Overall Risk Index), crop statistics, and recent case files. High-severity alerts appear as dynamic, animated banners on top, with actionable steps mapped directly on-screen.

• Weather Intelligence Center (Open-Meteo Integration)

A flagship weather engine that replaces static rain/temp displays with agricultural analysis. Connects to real-time **Open-Meteo forecasts** to fetch apparent temperatures, UV indexes, cloud covers, and sunrise/sunset times. Renders WMO symbol status cards for a 7-day outlook, tagging each day with a suitability rating (*Good for Farming*, *Use Caution*, or *High Risk*). Uses Gemini AI to calculate a 6-dimension risk assessment (Crop Stress, Disease, Waterlogging, Irrigation Need, Heatwave, and Pest Outbreak) along with personalized crop recommendations and protected impact estimates (water saved, yield protected, diseases prevented, and profit preserved).

• Google Sheets Published CSV Q&A; Syncing Integration

Enables agricultural agronomists to update the expert Q&A; knowledge base dynamically without code changes. The FastAPI backend contains a /api/assistant/sync-qa background task endpoint that pulls from a published Google Sheet CSV URL (configured via GOOGLE_SHEET_CSV_URL) and merges new rows into the local expert_kb.csv file, skipping duplicates and ensuring offline availability.

• Multilingual Voice Assistant & Server TTS

Empowers farmers of all literacy levels. Integrated with Web Speech recognition mapping Indic locales (en-IN, hi-IN, ml-IN, te-IN). Advisory readouts are generated entirely **server-side using gTTS (Google Translate TTS)**. The backend converts text into cached MP3 files (using SHA-256 content hashes to bypass redundant generation), which the frontend plays back using standard HTML5 Audio. This ensures identical, high-fidelity audio readouts across iOS, Android, Windows, and macOS without requiring local OS language packs.

• Gemini AI Personalized Assistant

A specialized chat interface connected to the farmer's soil context, history, and the expert Q&A; knowledge base. Implements strict Indic language enforcement rules (Hindi, Telugu, Malayalam, English) ensuring responses never switch mid-conversation. Provides a structured 7-day action timeline, confidence scores, and evidence sources for every agronomist recommendation.

• AI Crop Recommendation Engine

An intelligent crop selection module. Calibrates local weather, land size, and soil parameters against optimized crop profiles. Renders a confidence percentage match, a detailed yield breakdown, and risk factors. Includes a comparative matrix that showcases parameters alongside secondary crops to allow farmers to select the most profitable options.

• Multimodal Crop Disease Diagnoses (Gemini Vision)

Farmers upload leaves directly from fields. The frontend prompts for additional spoken descriptions (Whisper simulated processing). The server-side Gemini 2.5 Flash Vision model performs multimodal structural analysis, extracting symptoms (rust, leaf curling, spots) and generating immediate treatment recommendations (chemical names, biological controls). High-severity diagnoses are automatically escalated into CRM case tickets.

• CRM Agricultural Officer Command Console

A comprehensive management portal for government extension agronomists. Tracks district case queues (Pending, In Progress, Resolved) and allows officers to add remarks, prescribe treatments, and assign field visits. Includes a **Gemini District Copilot Center** that aggregates active complaints and alerts to generate automated government action checklists, weather vulnerability indexes, and prioritized farmer mobile outcalls.

• System Analytics Hub

Provides overall system health charts. Leverages Recharts to render crop distribution percentages, weather alert frequencies (Dry Spell, Heatwave, Storm), active case trends, and total yields protected, showing the platform's macro scalability.

• Guided Judge Walkthrough & Tech Showcase

Specially designed for hackathon evaluation. The Judge walkthrough offers a step-by-step evaluation workflow (Critical/Medium/Healthy profiles, weather impact triggers, Gemini Vision uploads, and officer audits). The Tech Showcase explains the multi-layered stack, offline-fallbacks, and database schemas under the hood.

3. Technical Architecture Specification

KisanVriddhi is engineered with a split client-server architecture designed for high availability, low latency, and sandboxed execution environment safety:

Backend API Architecture (FastAPI)

- **FastAPI Framework:** Asynchronous REST APIs with automatic OpenAPI (Swagger) schema validation.
- **Security Hardening:** Enforces strict security headers (nosniff, frame-ancestors, CSP, XSS protection). Implements Bearer token auth stored in secure HttpOnly cookies to prevent client-side token leaks.
- **Database (SQLite + SQLAlchemy):** Relation schemas mapping users, farmer profiles, alert logs, diagnosis cases, and chat records.
- **Gemini AI Integration:** Server-side wrapper connecting to Gemini 2.5 Flash (Text/Vision) with automatic failover fallback to Gemini 2.0 Flash to guarantee service continuity.
- **Server TTS Engine:** gTTS module producing high-quality MP3s served with Cache-Control headers. The audio endpoints are protected from path-traversal vulnerability attacks.

Frontend Architecture (React + Vite)

- **React (TypeScript):** Structured component tree separating dashboards, assistants, and centers.
- **Aesthetics & UI:** Premium glassmorphism effects, Tailwind CSS, Outfit typography, and dynamic framer-motion micro-animations.
- **HTML5 Audio:** Audio readouts streamed from `/api/tts/audio/{filename}` directly via standard browser media streams, removing OS voice dependencies.
- **Responsive Layout:** Mobile-first layouts optimized for Indian farmers using affordable android smartphones.

4. Feature & Platform Comparison

Feature Checklist	Traditional Agritech	KisanVriddhi Platform
Weather Advisories	Static 3-day forecast numbers	**AI Risk analysis + Open-Meteo Integration**
Indic Readouts	OS Voice-dependent (gibberish default)	**Server MP3 generation (All Devices)**
Pest & Disease	Manual identification checklists	**Gemini Vision Diagnostic + Escalation**
Expert Knowledge Base	Hardcoded static website resources	**Google Sheets Published CSV sync**
Officer Support	Static list of local offices	**Interactive CRM Case Ticketing & Briefs**
Indic Languages	English only text layouts	**EN, HI, ML, TE complete translation**